Hierarchical CNN Classification of Freshwater Zooplankton

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1 Introduction

The surveying of zooplankton populations is a key component of the Ontario Ministry of Natural Resources' (OMNR) fisheries monitoring program (Ontario Ministry of Natural Resources and Forestry 2024). Zooplankton monitoring in particular is essential for accessing the health of freshwater ecosystems. For instance, the spread of invasive dreissenid mussels in the Laurentian Great Lakes[1] over 2000-2019 has been attributed with zooplankton biomass reductions, contributing to declining whitefish populations (Ebener, Dunlop, and Muir 2021).

Traditional zooplankton surveying relies on manual classification of sampled zooplankton and non-zooplankton particles by trained taxonomists. More recent methods in machine learning, such as Convolutional Neural Networks (CNNs), offer a more efficient automated alternative. OMNR has explored several such methods, including a multi-class CNN and a hierarchical branch CNN (BCNN) for classifying images as zooplankton species or non-zooplankton objects, both achieving accuracies exceeding 95% on held-out test images (Ruiz Suarez and Haque 2025; Ruiz Suarez 2026).

This report evaluates two alternative hierarchical CNN architectures based on the Local Classifier Per Parent Node (LCPN) approach (Silla and Freitas 2011) against a flat CNN baseline for classifying images of freshwater zooplankton and non-zooplankton debris. We examine whether a taxonomic hierarchy, and a targeted extension of it, improves classification performance over the baseline, particularly for visually similar species such as Harpacticoid and Cyclopoid copepods.

2 Data and Data Processing

2.1 Dataset

The dataset used for model training and evaluation is a non-public collection of pre-labelled images of freshwater zooplankton and non-zooplantkon debris sampled from Ontraio lakes including Lakes Erie, Simcoe, and Huron. The images are taken using FlowCam devices, which capture high resolution microscopic images of particles from water samples (Yokogawa Fluid Imaging Technologies, n.d.), and are provided by the OMNR.

Raw images are provided as both mosiacs, single `.tif` files containing multiple labelled sample images, and as labelled `.tif` files of individual zooplankton and non-zooplankton samples. The labels of the provided images are encoded in their parent directories, e.g. images of bubbles are contained within a `bubbles/` directory. A Python script available from Ruiz Suarez (2026) was used to extract individual samples from the mosiac images.

[1] The Laurentian Great Lakes will be referred to as the "Great Lakes" hereafter.

Table 1: **Raw Image Directories and Corresponding Classes**

Class	Raw Directory	Image Count
Bosmina	bosminidae	2,912
Bosmina	eubosmina	280
Daphnia	daphnia	2,863
Rotifer	rotifer	2,056
Rotifer	trichocerca	1
Rotifer	conochilus	34
Rotifer	kellicottia	30
Nauplius	nauplius_copepod	4,708
Cyclopoid	cyclopoid	180,257
Harpacticoid	harpacticoid	1,713
Calanoid	calanoid	10,412
Exoskeleton	exoskeleton	1,123
Fiber	fiber_hairlike	5,288
Fiber	fiber_squiggly	6,455
Plant Matter	plant_matter	554
Bubbles	bubbles	4,834

Following image extraction from mosaics, the raw dataset contains 223,520 fully labelled images across 16 directories, which are consolidated into the 11 final classes used for model training (Table 1). The 11 final classes comprise the leaf nodes of a taxonomic hierarchy provided by OMNR, which organises freshwater zooplankton and non-zooplankton particles into a four-level tree structure (Figure 1).

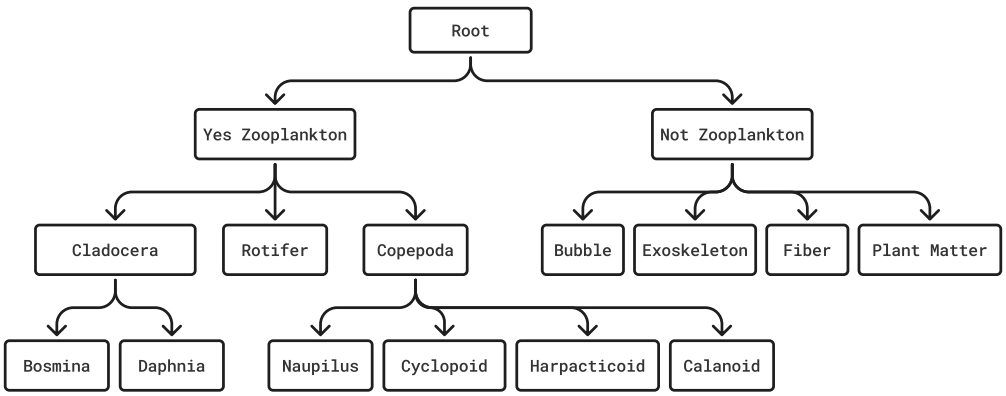

Figure 1: **Taxonomic Hierarchy of Freshwater Zooplankton.** Hierarchy provided by OMNR, organising zooplankton and non-zooplankton particles into a four-level tree structure. Leaf nodes correspond to the 11 classes used for model training and evaluation.

An image from each of the 11 leaf classes is shown below in Figure 2.

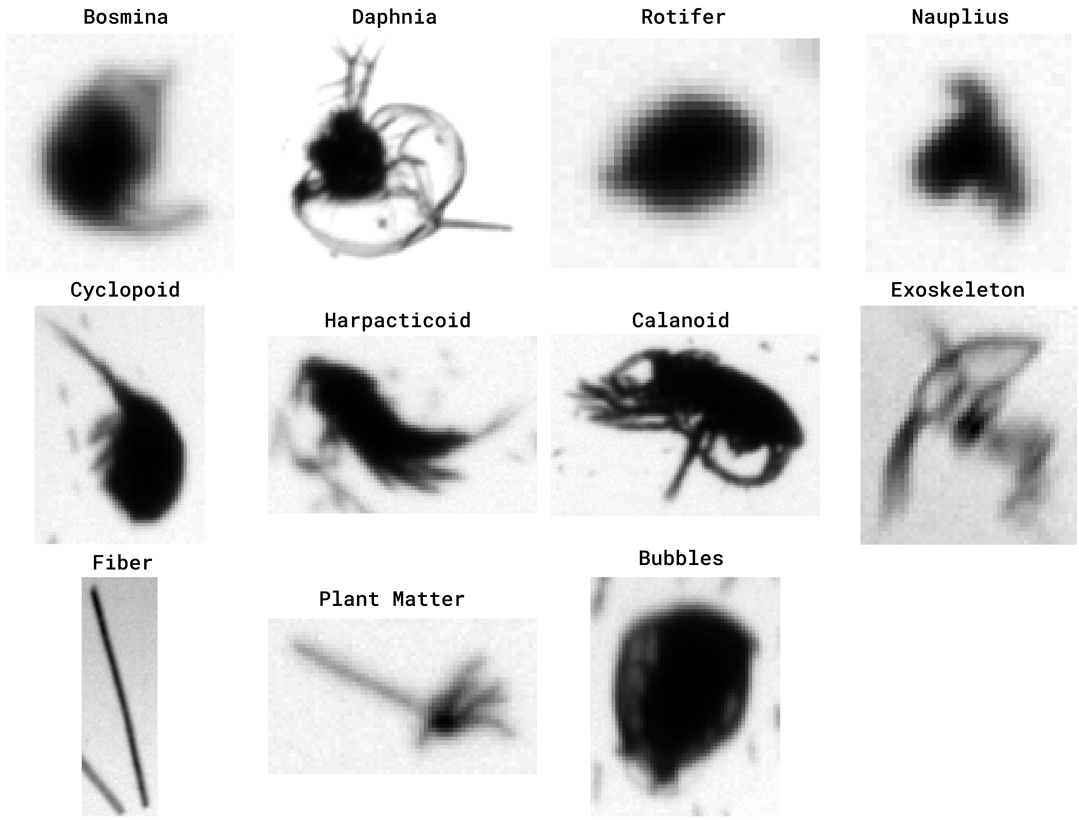

Figure 2: **Zooplankton and Non-Zooplankton Leaf Classes.** FlowCam images for each of the 11 leaf classes used for model training. The first 7 classes are species of zooplankton and the last 4 are non-zooplankton particles.

Additionally, the raw dataset contains two directories, `cladocera/` and `copepoda/`, containing partially labelled images corresponding to parent nodes on the hierarchy (Table 2).

Table 2: **Raw Image Directories and Corresponding Partial Classes**

Class	Raw Directory	Image Count
Cladocera	cladocera	776
Copepoda	copepoda	455

These images are included in the training of the hierarchical CNN models, but are excluded from the training of the flat baseline model and are excluded from the validation and testing of all models.

2.2 Preprocessing

Prior to model training and evaluation, all images are converted to greyscale on the basis that zooplankton identification relies primarily on morphological features, e.g. body shape, appendage

structure, and silhouette, rather than colour, consistent with how taxonomists manually classify zooplankton specimens (Leow et al. 2015). Additionally, conversion from 3 Red-Green-Blue (RGB) colour channels to a single greyscale channel reduces image size by two-thirds, lowering memory requirements and computational cost during training.

To improve model generalisation, three augmentations are applied randomly to training images: a horizontal flip, vertical flip, and random 90 degree rotation. These transformations are motivated by the lack of a canonical orientation of particles captured by FlowCam devices. Unlike images of a human, for instance, which typically show a person sitting or standing upright, zooplankton specimens submitted for classification may appear in any orientation. None of these augmentations are applied at validation or test time.

2.3 Data Splitting

Following the implementation of Ruiz Suarez (2026), each of the 11 leaf classes is capped at 10,000 images to prevent the over-representation of majority classes, following which the images are then divded into training, validation, and test sets using a stratified 70/10/20 split.

Per class sample caps are neccisary given the substantial class imbalance in the raw dataset, in which *Cyclopoid* images accounting for the majority of all images (180,257). Stratification ensures that the class distribution is preserved across all three splits and, in particular, ensures that minority classes are present in the testing dataset for evaluation purposes.

Table 3: **Training, Validation, and Test Split Counts for the Flat and LCPN Models**

Split	Flat	LCPN
Total	52,851	54,082
Train	36,992	37,853
Validate	5,282	5,404
Test	10,577	10,825

The flat and hierarchical CNNs are trained on slightly different sample sizes (Table 3). In particular, the flat model is trained on fully labelled leaf class images only, while the hierarchical models additionally incorporate the 1,231 partially labelled images assigned to the *Cladocera* and *Copepoda* nodes of class hierarchy (see Figure 1). Figure 3 shows the training, validation, and testing counts per class after sample sizes are capped.

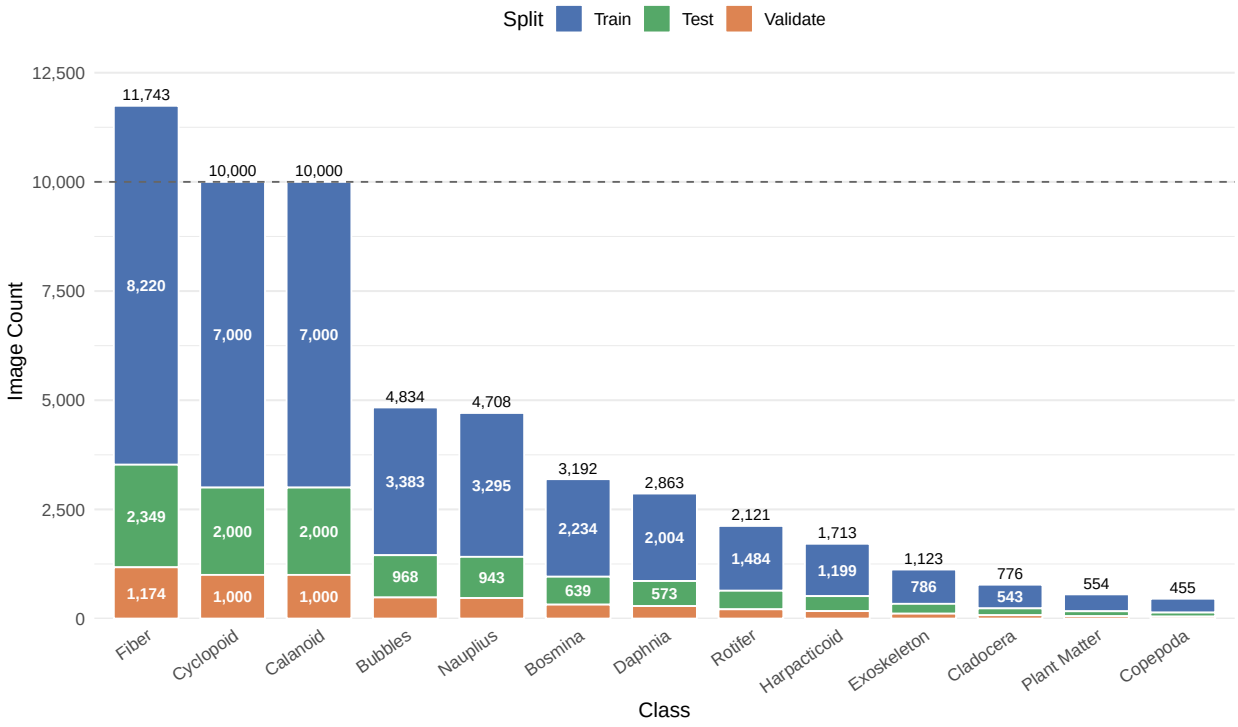

Figure 3: **Class Distribution After Capping.** Image counts per class after applying a 10,000 image cap. Includes the partially labelled image classes Cladocera and Copepoda. Bars are divided into training (70%), validation (10%), and test (20%) splits. Labels within each segment show the split count where space permits (minimum 500 images) and labels above each bar show the total class count. See Table 4 for the complete counts. The dashed line indicates the 10,000 image cap.

Note that due to a bug in the image loading implementation, images of the *Fiber* class were not capped at 10,000, resulting in 11,743 images being used for training rather than the intended 10,000. This discrepancy did not materially affect model performance, and the uncapped total results in *Fiber* being only slightly over-represented relative to the target cap. As the classification of non-zooplankton objects is not a primary focus of this investigation, the models were not re-trained following the discovery of this bug.

Appendix

A Additional Tables and Figures

Table 4: **Complete Class Distribution After Capping**

Class	Raw Count	Capped Count	Train	Validate	Test
Bosmina	3,192	3,192	2,234	319	639
Daphnia	2,863	2,863	2,004	286	573
Rotifer	2,121	2,121	1,484	212	425
Nauplius	4,708	4,708	3,295	470	943

Table 4: **Complete Class Distribution After Capping**

Class	Raw Count	Capped Count	Train	Validate	Test
Cyclopoid	180,257	10,000	7,000	1,000	2,000
Harpacticoid	1,713	1,713	1,199	171	343
Calanoid	10,412	10,000	7,000	1,000	2,000
Exoskeleton	1,123	1,123	786	112	225
Fiber	11,743	11,743	8,220	1,174	2,349
Plant Matter	554	554	387	55	112
Bubbles	4,834	4,834	3,383	483	968
Cladocera	776	776	543	77	156
Copepoda	455	455	318	45	92

References

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